



VERITAS UNIVERSITY, ABUJA
Bwari Area Council, FCT
Website: <http://www.veritas.edu.ng>
FIRST SEMESTER EXAMINATION

Faculty: Natural And Applied Sciences Department: Software Engineering Session: 2025/2026
Course Code: VUNA- SEN 181 Course Title: SOFTWARE ENGINEERING LAB I (Java Programming Practical)
Units: 2 Duration: 3 Hours Date: 02/03/2026 Venue: SEN LAB
Instructions: **Answer Question 1 And Any Other 2 Questions
**SAVE YOUR WORK IN A FOLDER ON THE DESKTOP USING YOUR MATRIC NUMBER AND NAME

QUESTION 1 ✓

Election Eligibility Checker and Voting System

Write a program for an election exercise that accepts: Age, Citizenship, and political party(A, B, and C) for 10 users.
A user is eligible if:

Age is ≥ 18 AND ≤ 80

Citizenship equals "Nigerian"

Display whether the user is Eligible or Not Eligible.

If Eligible, the user can vote for only one party (A, B, or C).

The system keeps count of: (i.) How many are eligible or not.

(iii.) the winner of the election.

(ii.) how many voted for each party and

QUESTION 2

a. Login Verification System

Write a Java program that uses a username "admin" and password "java123". Allow the user three (3) attempts to log in. If login is successful, print "Access granted". But If login is not successful, print "Access denied, please try again". After three failures, print "Account locked". and end the program.

b. Write a program to print the combination of numbers. Where $n! = n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 3 \times 2 \times 1$

$${}^nCr = \frac{n!}{(n-r)!r!}$$

QUESTION 3

Student Result Processing System

Write a Java program that:

Uses an array to store the scores of 8 courses.

Uses a loop to compute:

Total score

Average score

Displays:

Highest score and the course

Lowest score and the course

Gives a list of how many courses scored:

Above average (average=50)

Below average (average=50)

QUESTION 4

a. Write a program that:

Accepts a number.

If it is divisible by 3 and 5, print "FizzBuzz".

If divisible by only 3, print "Fizz".

If divisible by only 5, print "Buzz".

Otherwise, print the number.

b. Write a java program to produce the following sequence of numbers to the 50th element

Fibonacci series: - 1, 1, 2, 3, 5, 8, 13, 21, 34,

1+1=2, 1+2=3; 2+3=5; 3+5=8; 5+8=13

QUESTION 5

Write a program that contains:

A method called calculateArea() that receives the radius of a circle and returns the area. Note: -

Area=Pi*radius*radius

A method called calculatePerimeter() that returns the perimeter. Note: - Perimeter=2*Pi*radius.

Where Pi=3.17

Call both methods inside main() and display the results